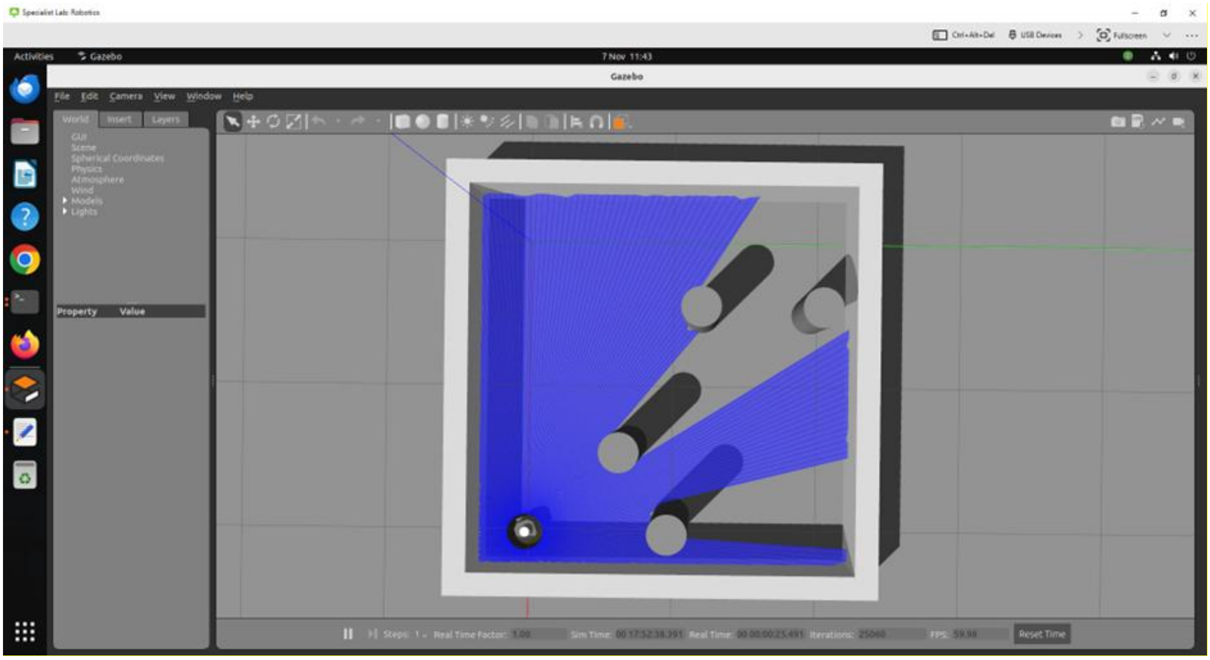
Lab 6 Continue PID work and Optimize your laser code

In this lab, you should continue working on your PID control code and then improve your laser code by following the steps below.

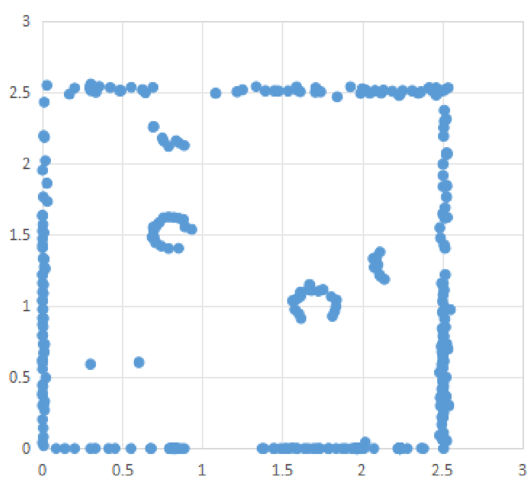
6.1 Continue PID based control coding

Firstly, make sure your code can control the robot to pass two gaps without collision and reach the charger accurately and robustly.

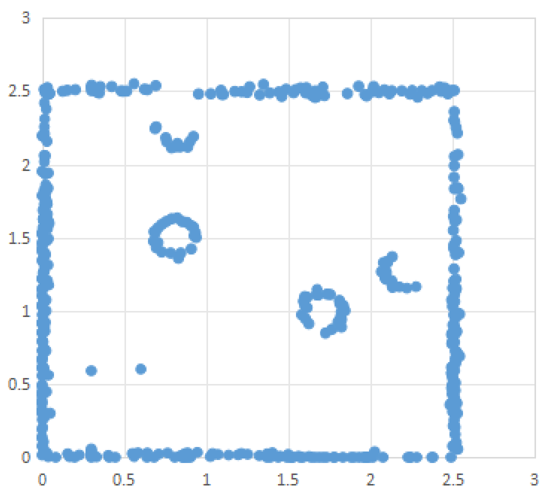


6.2 Map improvement

- Revising the laser code to improve the quality of the environment map.
- Plotting a new graph with the robot trajectory and the improved map.
- Briefly explain your new laser code using flowchart and text.



The map before improvement



The map after improvement

6.3 Optimize your code

- You should remove any piece of code that is not used for implementing the challenge task. In this way, you can shorten and optimize your code.
- You may reduce the redundancy code by combining them into a common function such that your code can be effectively shortened and in a modular way. For example, the code for collecting laser data and converting them into global coordinates can be simplified.
- Add more comments to make your code easy to understand, such as

  case 6:
  moveStop();   // to stop the robot at the charger position.

- Finally, you should avoid using digital numbers in your main code by defining some meaningful variables to replace digital numbers, such as

  laser_landmark3 = 0.6

  in this way, *laser_landmark3* is used in your main code as follows:

  if (PositionY < odom_landmark3)

  instead of *0.6*.

  if (PositionY < 0.6)

- Print all the graphs to be included into your assignment report. Note the units cannot be missed.